Guanhao Wang

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Portfolio · GitHub
Computer science undergraduate focused on game level design, environmental storytelling, game interaction, and physical computing.

Education

University of Wisconsin–Madison | Computer Science Undergraduate **June 2025 to present**
Expected graduation May 2027 · GPA 3.724/4.000

The Pennsylvania State University | Undergraduate Study in Computer Science **August 2023 to June 2025**
GPA 3.94/4.00 · Transferred to UW–Madison

Games and Interactive Projects

Coal Echoes | Narrative Exploration Game **2026**
In development · Unity 6, HDRP, C#, Blender

- Developing a first person exploration game inspired by family memories of Shitanjing, Ningxia, using photography, a grandfather' s diary, and scrapbooking to tell a story about an abandoned coal plant.
- Built photography, album, scrapbook layout, object inspection, and echo discovery systems, with local storage for photographs, layouts, diary progress, and discovered echoes.
- Integrated DEM terrain, industrial buildings, and a moving cableway, and developed sandstorm, heat haze, and newspaper photo transition effects to support the setting.

Source code on GitHub

Project Fireflies | Interactive Light Installation **2026**
Completed · ESP32, Arduino, Unity, UDP

- Mapped movement and ambient light inputs to firefly disturbance and synchronization behavior using LD2410B radar, light sensors, and WS2812B LEDs.
- Configured firmware for 420 LEDs across three swarms, with two independently responsive zones, one ambient swarm, and a UDP interface for Unity synchronization commands.

Seafloor Revival | GMTK Game Jam 2025 **July to August 2025**
UI Development and Art Integration · Unity, C#

- Built all game UI in a team of six credited creators, wired component references and related C# behavior, and processed and integrated interface art assets.
- Contributed to a 2.5D underwater base building game about recurring pollution disasters, released as a playable browser prototype.

Playable browser prototype

Additional Projects and Research

Mask | Global Game Jam 2026 **2026**
Team Game Jam Entry · Unity, C#

- Participated in a 2D narrative platformer about identity and social roles. The project combines household tasks, branching phone conversations during a subway commute, and memories of different identities.

Game Jam entry

Online Computer Graphics Project | Mooncake **February to April 2025**
Course led by Prof. Saty Raghavachary · Three.js, JavaScript

- Studied 3D modeling, rendering, animation, and GUI interaction; developed the particle system and web demonstration with controls for particle count and size.
- Implemented interactive controls with Three.js and lil GUI and documented the graphics implementation and technical approach in a project report.

Project demonstration

Human Computer Interaction in Indie Games | Published Paper **Research November 2022 to June 2023**
Published 2023 · Sole Author

- Analyzed FAR Lone Sails through its official artbook and gameplay, examining implicit guidance, environmental storytelling, mood building, and emotional attachment to a vehicle.
- Published in Lecture Notes in Education Psychology and Public Media, volume 18, pages 194 to 199.

Publication and DOI

Professional Experience

IQVIA | Technology Solutions Intern **July to August 2024**
Shanghai, China

- Created and maintained over 100 pages of user documentation for a pharmaceutical company's WeChat Mini Program platform upgrade, covering two user types, four scenarios, and three device contexts.
- Performed user acceptance testing, reported over 10 defects through Jira, and explained reproduction steps and cross platform scenarios in project meetings.
- Organized over 1,000 data tags into a three level hierarchy tied to business scenarios to support user filtering and tag prioritization.

Skills and Languages

Games and interaction Unity, C#, HDRP, uGUI, TextMesh Pro, Shader Graph, Blender, ESP32, Arduino
Programming and tools Python, Java, C, JavaScript, Three.js, Git, Git LFS, Jira
Languages Mandarin native, English fluent, Japanese basic